

Bit Inversions

Time limit: 1.00 s

Memory limit: 512 MB

There is a bit string consisting of n bits. Then, there are some changes that invert one given bit. Your task is to report, after each change, the length of the longest substring whose each bit is the same.

Input

The first input line has a bit string consisting of n bits. The bits are numbered $1, 2, \dots, n$.

The next line contains an integer m : the number of changes.

The last line contains m integers $x_1, x_2, \dots, x_m$ describing the changes.

Output

After each change, print the length of the longest substring whose each bit is the same.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq x_i \leq n$

Example

Input	Output
001011 3 3 2 5	4 2 3

Explanation: The bit string first becomes 000011, then 010011, and finally 010001.